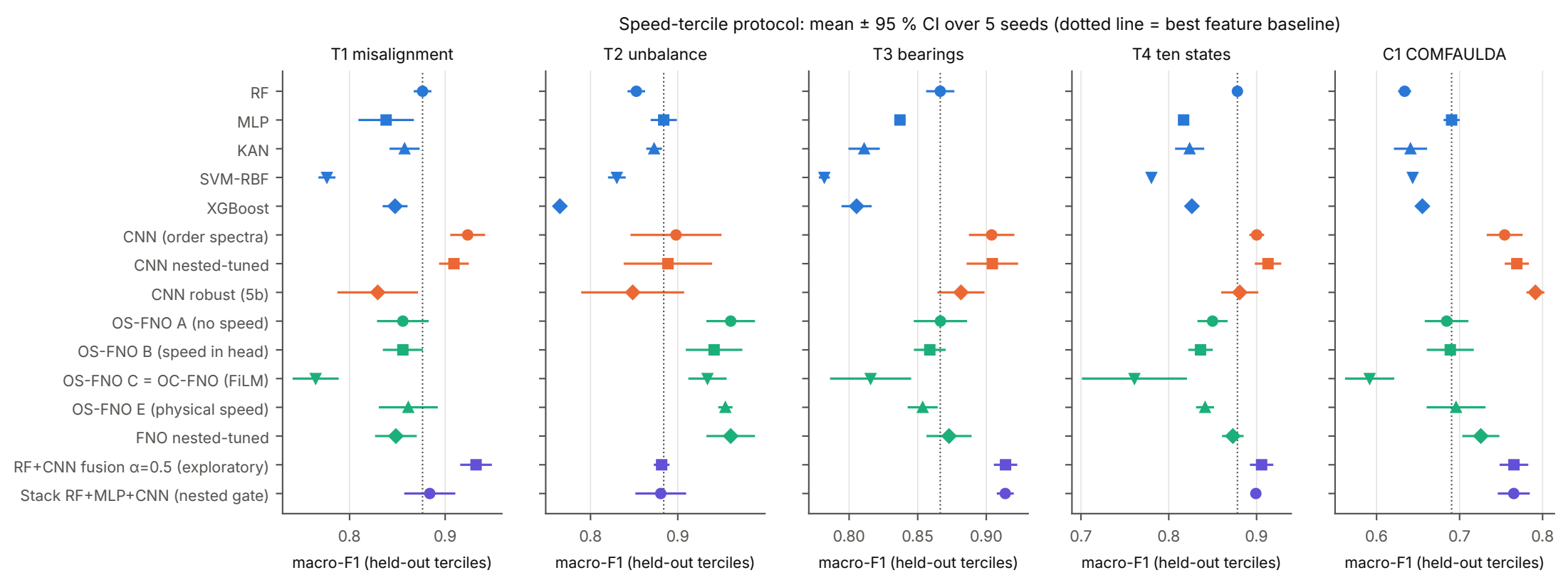
Speed-tercile protocol: mean ± 95 % CI over 5 seeds (dotted line = best feature baseline)
T1 misalignment
T2 unbalance
T3 bearings
T4 ten states
C1 COMFAULDA
RF
MLP
KAN
SVM-RBF
XGBoost
CNN (order spectra)
CNN nested-tuned
CNN robust (5b)
OS-FNO A (no speed)
OS-FNO B (speed in head)
OS-FNO C = OC-FNO (FiLM)
OS-FNO E (physical speed)
FNO nested-tuned
RF+CNN fusion α=0.5 (exploratory)
Stack RF+MLP+CNN (nested gate)
0.8
0.9
0.80
0.85
0.90
0.7
0.6
macro-F1 (held-out terciles)